A Fractional-Divisor Obstruction for a Normalized Hénon Determinant

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Abstract

The field-degree-normalized Galois norm of a finite-field Hénon determinant defines a canonical Euler germ on $\operatorname{Re} s > 1/2$. We test whether its local roots are ordinary rational determinant ratios. At the first split prime $p = 7$, exact cyclotomic sector arithmetic and real-subfield resultants give $N_7 = P_{18}^2/(49P_{12}^2)$, where P_{18} and P_{12} are coprime and squarefree. Since the normalizing field degree is three, the local root has orders $\pm 2/3$. It is neither rational nor single-valued meromorphic across its unit-circle divisor and cannot be an ordinary finite-dimensional graded determinant. The critical-boundary germ itself survives. The result forces a precise operator-category pivot: any determinant realization must use a normalized trace admitting fractional projection dimensions.

1 Introduction

An analytic logarithm may define a perfectly canonical germ even when its exponential has fractional divisor orders after continuation. This distinction becomes unavoidable in the Hénon Galois descent constructed in the preceding stage. For each split prime, the complete rational norm $N_p(z)$ is divided logarithmically by $d_p = [\mathbf{Q}(\zeta_p)^+ : \mathbf{Q}]$. The resulting Euler product reaches $\operatorname{Re} s > 1/2$, but it is an ordinary determinant only if every valuation in $\operatorname{div} N_p$ is divisible by d_p.

We settle that gate negatively at $p = 7$.

Theorem 1.1 (Fractional-divisor obstruction)**.** *Let N_7 be the rational Galois norm of the conjugate-paired μ_3 Hénon augmentation factor. Then*

$$N_7(z) = \frac{P_{18}(z)^2}{49P_{12}(z)^2},$$

where P_{18} and P_{12} are squarefree and coprime in $\mathbf{Q}[z]$. Hence N_7 is not a cube, and the origin branch

$$G_7(z) = \exp\left(\frac{1}{3} \operatorname{Log}_0 N_7(z)\right)$$

has local orders $\pm 2/3$. It does not extend through any local divisor point as a single-valued meromorphic scalar function.

The theorem does not refute the half-plane germ. It identifies the precise reason an ordinary determinant is the wrong category. Normalized traces in finite or semifinite algebras can assign fractional dimensions to projections; C47 tests whether that mechanism realizes the full Hénon germ.

2 Exact sector descent

Let $\zeta = \zeta_7$, choose $\rho = 2$, and let $T = U_7^2$. For the three eigenspaces H_k of the coordinate permutation, set

$$D_k(z) = \det(I - zT \mid H_k).$$

Exact arithmetic in $\mathbf{Q}(\zeta)$ gives $D_2 = D_1$. Let $\theta = \zeta + \zeta^{-1}$, with

$$m(\theta) = \theta^3 + \theta^2 - 2\theta - 1.$$

The conjugate-paired sector factors

$$q_k(\theta, z) = D_k(z)\overline{D_k(z)}$$

belong to $\mathbf{Q}(\theta)[z]$. Their explicit reductions are recorded in Appendix A. Since the augmentation factor is $(D_0/D_1)^2$, the paired factor equals $(q_0/q_1)^2$.

All operations retain the two-step kernel chronology. Conjugation pairs the additive character; it does not average a transition matrix or alter the prime clock.

3 Resultants and divisor multiplicities

Because m is monic, taking the norm from $\mathbf{Q}(\theta)$ is equivalent to a resultant in θ. Exact elimination gives

$$\mathrm{Res}_\theta(m, q_0) = P_{18}/49, \qquad \mathrm{Res}_\theta(m, q_1) = P_{12}/7.$$

Consequently

$$N_7(z) = \frac{P_{18}(z)^2}{49 P_{12}(z)^2}, \qquad N_7(0) = 1. \tag{1}$$

The numerator and denominator of (1) have degrees 36 and 24. Their difference 12 agrees with the general ordinary-norm formula $2(p-1)$.

Proposition 3.1 (Exact gcd certificate)**.** *The polynomials P_{18} and P_{12} are squarefree and mutually coprime over $\mathbf{Q}$.*

Proof. Reduction modulo five preserves both degrees. Direct Euclidean algorithms in $\mathbf{F}_5[z]$ give

$$\gcd(P_{18}, P_{12}) = gcd(P_{18}, P_{18}') = \gcd(P_{12}, P_{12}') = 1.$$

A common factor or repeated factor over $\mathbf{Q}$ would give one after reduction at a degree-preserving good prime. Thus the three rational gcds are one. □

Every zero of N_7 now has valuation two and every pole valuation minus two. The points lie on $|z| = 1$: before Galois descent the factors are characteristic polynomials of unitary sector blocks, and Proposition 3.1 excludes cross-sector cancellation.

4 Proof of the branch theorem

Here $d_7 = 3$. A cube in $\mathbf{C}(z)$ has divisor valuations divisible by three. The exact valuations ± 2 supplied by Proposition 3.1 prove that N_7 is not a cube. Near a zero α,

$$G_7(z) = (z - \alpha)^{2/3} h(z), \qquad h(\alpha) \neq 0,$$

and a pole gives order $-2/3$. A single-valued meromorphic function has integral local order. This proves Theorem 1.1.

The obstruction is stronger than the statement that a convenient rational formula was not found. It is invariant under rational re-expression of the norm and is already present at the first split prime. Any ordinary finite-dimensional determinant ratio has integral local orders and therefore cannot equal G_7.

It would be incorrect, however, to infer a global natural boundary. The origin branch remains holomorphic on $|z| < 1$, and the C45 prime product remains holomorphic on $\operatorname{Re} s > 1/2$. Cross-prime monodromy, global cancellation, and alternative operator categories have not been classified.

5 Route-A decision

The C45 analytic germ and the source-native two-step quantization survive. The ordinary determinant promotion fails exactly, and there is no functional equation, Gamma factor, or continued scalar divisor. The frozen evaluation is

$$\begin{aligned} &\mathrm{A1} = \mathrm{WEAK}, && \mathrm{A2} = \mathrm{ANALYTIC\_DETERMINANT}, \\ &\mathrm{A3} = \mathrm{FAIL}, && \mathrm{A4} = \mathrm{NATURAL\_QUANTIZATION}. \end{aligned}$$

Overall: `STOP_ORDINARY_DETERMINANT_PROMOTION`. Route B is not authorized.

The conclusion is useful rather than merely negative. Fractional orders are exactly what a normalized trace on a finite algebra can encode. C47 therefore keeps the Hénon dynamics fixed and changes only the determinant category, testing whether a legitimate graded normalized-trace determinant reproduces the C45 germ.

A Frozen polynomials

The real paired sector reductions are

$$\begin{aligned} 7q_0 = &-15\theta^2 z^5 - 26\theta^2 z^4 - 26\theta^2 z^3 - 26\theta^2 z^2 - 15\theta^2 z \\ &- 3\theta z^5 - 10\theta z^4 - 19\theta z^3 - 10\theta z^2 - 3\theta z \\ &+ 7z^6 + 31z^5 + 61z^4 + 72z^3 + 61z^2 + 31z + 7, \\ 7q_1 = &\, 4\theta^2 z^2 + 5\theta z^2 + 7z^4 - 12z^2 + 7. \end{aligned}$$

The norm polynomials are

$$\begin{aligned} P_{18}(z) = &\, 49z^{18} + 147z^{17} + 147z^{16} - 14z^{15} - 133z^{14} - 63z^{13} + 71z^{12} \\ &+ 104z^{11} + 50z^{10} + 13z^9 + 50z^8 + 104z^7 + 71z^6 \\ &- 63z^5 - 133z^4 - 14z^3 + 147z^2 + 147z + 49, \\ P_{12}(z) = &\, 7z^{12} - 21z^{10} + 35z^8 - 41z^6 + 35z^4 - 21z^2 + 7. \end{aligned}$$